

Chapter 6. The Normal Probability Distribution

Table of Contents

6.1 Properties of the Normal Distribution	2
Uniform Distribution	2
The Normal Distribution.....	4
Properties of the Normal Curve	5
Area Under a Normal Curve	8
6.2 Applications of the Normal Distribution	10
Standardizing a Normal Random Variable	10
Finding Areas Under the Normal Curve:.....	11
Find the Value of a Normal Random Variable	14
6.3 Assessing Normality	15
Using a Frequency Histogram:	15
Using a Normal Probability Plot:.....	16

By the end of this chapter, students should be able to:

- Describe a uniform distribution and use it for probability calculations.
- Describe properties a normal distribution
- Use the normal distribution for probability calculations.
- Assess if data is well approximated by a normal distribution.

6.1 Properties of the Normal Distribution

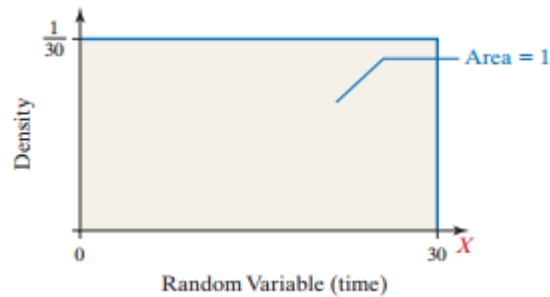
- In chapter 5, we talked about discrete random variables, where a random variable, X , had a finite or countable number of outcomes.
- In chapter 6, we will talk about continuous random variables, where a random variable had an infinite number of outcomes.

Uniform Distribution

- Imagine your friend is always running late. Let the random variable X represent the time from when you are supposed to meet your friend until they show up. Suppose your friend could be on time ($x = 0$) or up to 30 minutes late ($x = 30$), with all equal intervals of time between $x = 0$ and $x = 30$ being equally likely.
 - This means that your friend is just as likely to be 3-4 minutes late as they are to be 25-26 minutes late.
 - Since any two intervals of equal length between 0 and 30, inclusive, are equally likely, the random variable X is said to follow a **uniform probability distribution**.
- For a discrete random variable, the value of a random variable is usually substituted into a formula or table. For continuous variables, it's not as easy.
- For example, what is the probability that your friend is *exactly* 12.898746541 minutes late?

Definition: A **probability density function (pdf)** is

- The figure below shows a very basic probability density function.



- We can use area under the curve to represent the probability of observing a value of the random variable in that interval.

Example 1: Refer to the situation listed above.

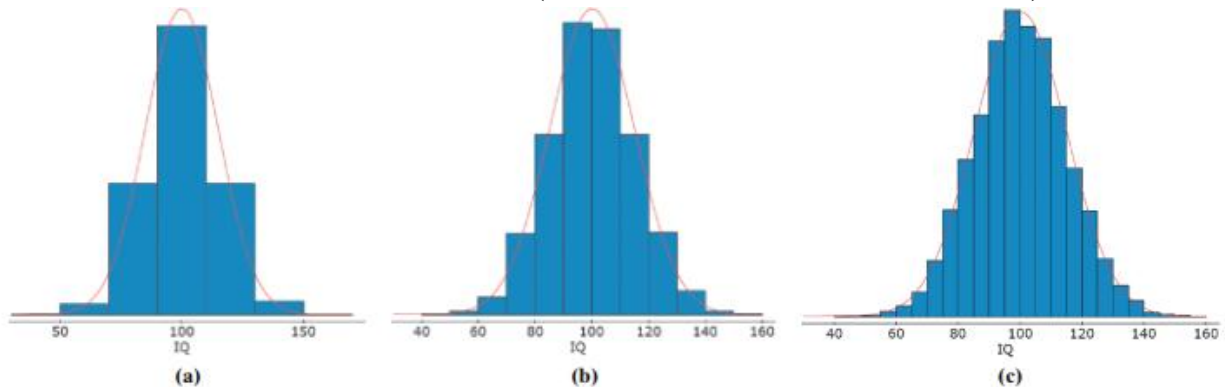
- What is the probability that your friend will be between 10 and 20 minutes late?
- You are scheduled to meet at 10am with your friend. Suppose it is 10am right now. There is a 20% probability that your friend will arrive within the next _____ minutes.

Example 2: Computers can create random numbers often called a random number generator (note that with computers, it's often seen as "pseudo" random as often times it's not truly random). However, suppose you randomly generate a number between 0 and 10.

- What is the probability that number you select is between 2 and 5?
- What is the probability that the number you select is exactly 4.35?

The Normal Distribution

- Not too many random variables follow a uniform distribution, but many random variables follow a bell-shaped distribution.
 - For example, IQ scores and birth weights of babies are commonly bell-shaped.
- If we took a random sample of 10,000 adults, we could create a histogram of IQ scores. As we increase the sample size, the histogram becomes more and more close to a smooth curve (as seen with the red curve below).

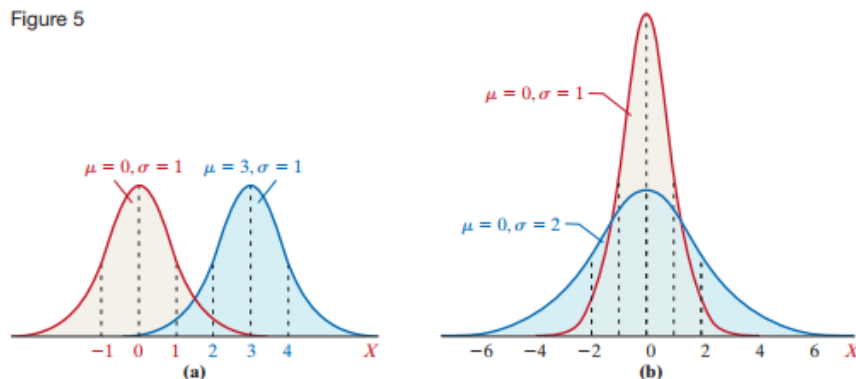


- We can use this red curve to model the probability distribution of a continuous random variable.

Definition: A continuous random variable is **normally distributed**, or has a **normal probability distribution**,

- Any normally distributed random variable follows a bell-shaped curve like above, but how does each distribution differ?

Figure 5

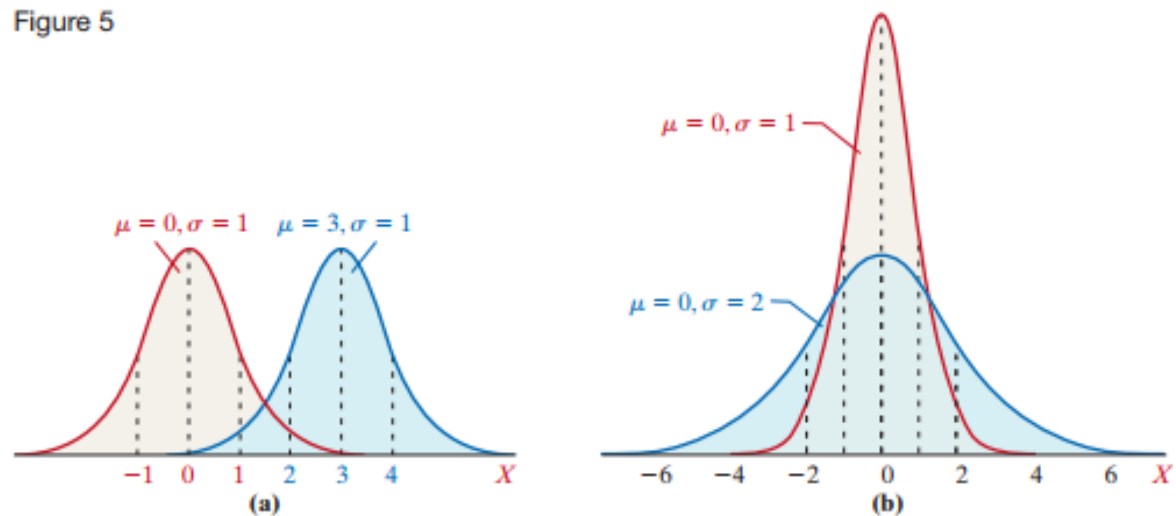


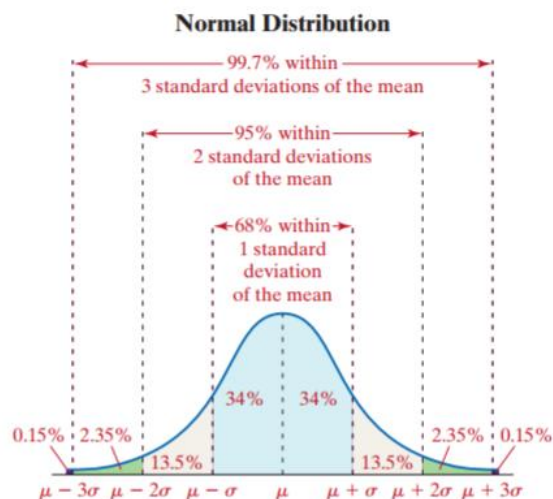
Properties of the Normal Curve

- To define a normal distribution, we need to know the mean and standard deviation of the random variable. Using those, we can list the properties of the distribution:

- 1.
- 2.
- 3.
- 4.
- 5.
- 6.
- 7.

Figure 5

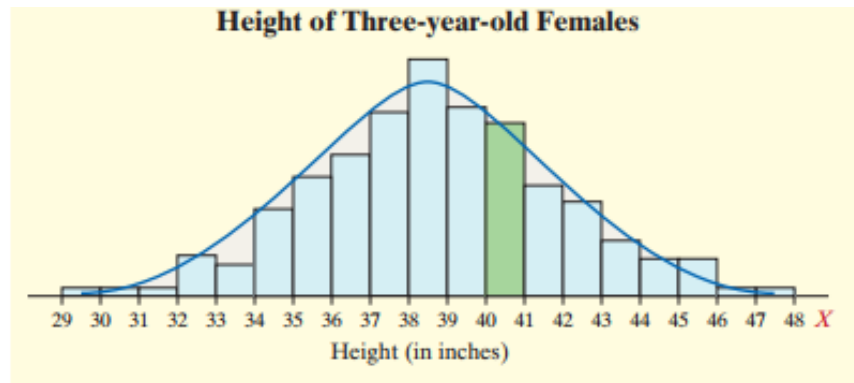
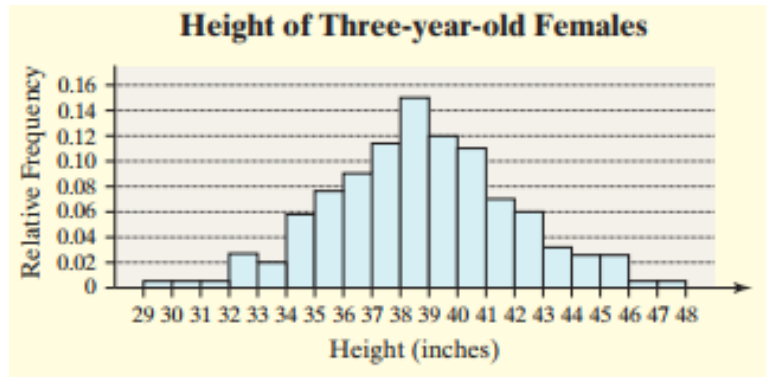




Example 3: The relative frequency distribution given in the table below represents heights of a pediatrician's three-year-old female patients. The raw data indicate that the mean height is $\mu = 38.72$ inches with a standard deviation of $\sigma = 3.17$ inches.

Height (inches)	Relative Frequency
29.0–29.9	0.005
30.0–30.9	0.005
31.0–31.9	0.005
32.0–32.9	0.025
33.0–33.9	0.02
34.0–34.9	0.055
35.0–35.9	0.075
36.0–36.9	0.09
37.0–37.9	0.115
38.0–38.9	0.15
39.0–39.9	0.12
40.0–40.9	0.11
41.0–41.9	0.07
42.0–42.9	0.06
43.0–43.9	0.035
44.0–44.9	0.025
45.0–45.9	0.025
46.0–46.9	0.005
47.0–47.9	0.005

- Look at the relative frequency histogram on the next page. Comment on the shape of the distribution.
- Look at the normal curve with $\mu = 38.72$ and $\sigma = 3.17$ on the relative frequency histogram. Does it appear that this model fits the data well?



- How do we create that curve to graph?
- The probability density function of a normal random variable is given by the following formula:

$$y = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

- Don't worry too much about the equation. We won't physically use it, but methods we cover have used this for us.

Area Under a Normal Curve

- Suppose that a random variable X is normally distributed with mean μ and standard deviation σ . The area under the normal curve for any interval of values of the random variable X represents either:

1.

2.

Note: Since $P(X = a) = 0$ for all values of a , then the following also holds:

$$P(a < X < b) = P(a \leq X < b) = P(a < X \leq b) = P(a \leq X \leq b)$$

Example 4: The serum total cholesterol for males 20-29 years old is approximately normally distributed with mean $\mu = 180$ mg/dL and $\sigma = 36.2$ mg/dL.

- (a) Sketch out a normal curve with the parameters labeled.
- (b) An individual with 200 mg/dL is considered to have high cholesterol. Shade in the region under the normal curve to the right of $x = 200$.
- (c) Suppose that the area under the normal curve to the right of $x = 200$ is 0.2903. Give two interpretations of this result.

Example 5: The length of human pregnancies is approximately normally distributed with a mean of 266 days and a standard deviation of 16 days.

- (a) Sketch out a normal curve with the parameters labeled.
- (b) An individual whose birth lasts 259 days or less is considered preterm.
Shade in the region under the normal curve that represents the proportion of births that are preterm.
- (c) The area to the left of $x = 259$ is equal to .3309. Give two interpretations of this result.

6.2 Applications of the Normal Distribution

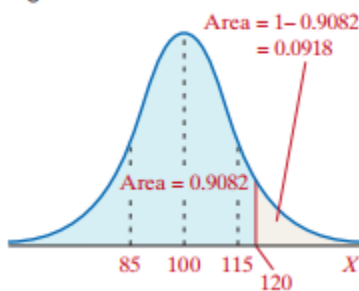
- If X is a normally distributed random variable, the area under the normal curve represents the proportion of a population with a certain characteristic.
 - How do we find the area under the normal curve?
- We have two options: by hand with tables or with technology (Excel).

Standardizing a Normal Random Variable

- **Definition:** Suppose that the random variable X is normally distributed with mean μ and standard deviation σ . Then the random variable:

- This result shows us that any normal random variable X can be transformed into a standard normal random variable Z whose mean is 0 and standard deviation is 1.
- The means that we can use a table to find the area to the left of a specified z -score.
 - For example, IQ scores can be modeled by a normal distribution with $\mu = 100$ and $\sigma = 15$. An individual whose IQ is 120:

Figure 12



Finding Areas Under the Normal Curve:

By-Hand Approach

Step 1: Draw a normal curve and shade the desired area.

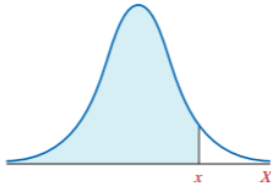
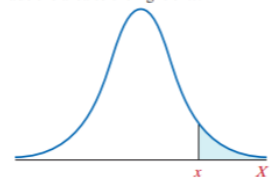
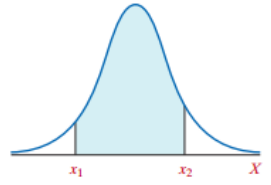
Step 2: Convert the value of x to a z -score using:

$$z = \frac{x - \mu}{\sigma}$$

Step 3: Use Table V in the textbook (or at the end of these notes) to find the area to the left of the z -score.

Example 6: A pediatrician obtains the heights of her three-year-old female patients. The heights are approximately normally distributed, with mean 38.72 inches and standard deviation 3.17 inches. Use the normal model to determine the proportion of three-year-old females who are shorter than 35 inches.

- Not all problems will involve areas to the left. How do we find other areas?

Problem	Solution (by hand)
Find the area to the left of x. Shade the area to the left of x. 	Convert the value of x to a z-score. Use Table V to find the area to the left.
Find the area to the right of x. Shade the area to the right of x. 	Convert the value of x to a z-score. Use Table V to find the area to the left. Find the final answer by doing 1 minus the area to the left.
Find the area between x_1 and x_2. Shade the area between x_1 and x_2. 	Convert the values of x to z-scores. Find the area to the left of them both (called z_1 and z_2). Final answer is (area to the left of z_2) – (area to the left of z_1).

Example 7: The scores earned on the mathematics portion of the SAT, a college entrance exam, are approximately normally distributed with a mean of 516 and standard deviation of 116. What is the probability that a randomly selected test taker would earn:

- (a) Less than 500 points?
- (b) More than 600 points?
- (c) Between 525 and 550 points?

Find the Value of a Normal Random Variable

- What happens if we have an area under the curve, but we want to find the value associated with it?

By-Hand Approach
Step 1: Draw a normal curve and shade the desired area.
Step 2: Use Table V to find the z-score that corresponds to the shaded area. You must convert this area to the area to the left of this z-score.
Step 3: Obtain the normal value using the formula $x = \mu + z\sigma$

Example 8: The heights of a pediatrician's three-year-old female patients are approximately normally distributed, with mean 38.72 inches and standard deviation 3.17 inches.

- Find the height of a three-year-old female at the 20th percentile.
- Find the heights that separate the innermost 80% of heights from the bottom and top 10% of heights.

6.3 Assessing Normality

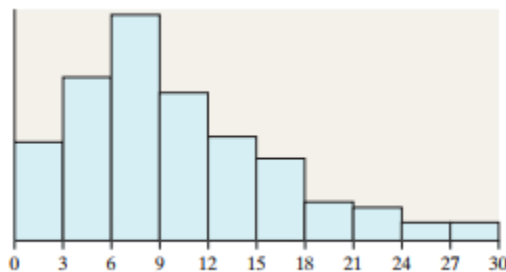
- In the real world, many variables (but not all) are bell-shaped. If you are provided with data, how can you assess normality?
- We have 2 basic methods to determine this:
 - Look at a frequency histogram.
 - Look at a normal probability plot.

Using a Frequency Histogram:

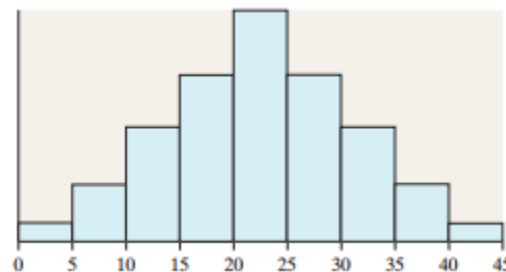
- This is the most straight-forward way to assess if a random variable can be well approximated by a normal curve.
- Generally, we want to see the following in a plot:
 - Bell shaped with a single peak.
 - The histogram should be roughly symmetric (with no strong skews).
 - There should be few outliers.

Example 11: Consider the wait times at two different restaurants. Consider the frequency distributions of the wait times below:

Restaurant A:



Restaurant B:

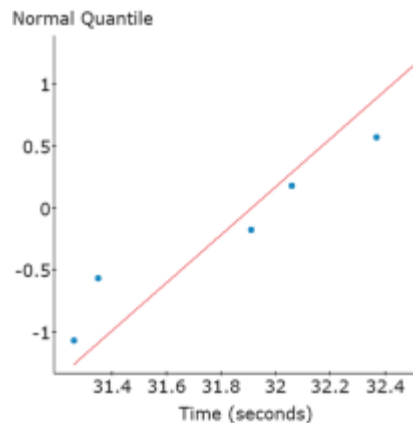


Which restaurant's wait time could be modeled by a normal distribution, if any?

Using a Normal Probability Plot:

- This plot is very common, especially if you go into higher level statistics courses.
- In a normal probability plot, we compare two things:
 - The actual value of the observations.
 - The expected z-score if the data was normally distributed.
- An ideal normal plot would have actual values perfectly associated with the expected z-scores, so they should create a straight line.

Example 12: Consider the table below of finishing time for six randomly selected races of a greyhound. Does it appear that finishing time is normally distributed?



Area under the curve (Table V):

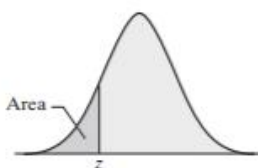


Table V										
Standard Normal Distribution										
z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
-3.4	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0002
-3.3	0.0005	0.0005	0.0005	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0003
-3.2	0.0007	0.0007	0.0006	0.0006	0.0006	0.0006	0.0006	0.0005	0.0005	0.0005
-3.1	0.0010	0.0009	0.0009	0.0009	0.0008	0.0008	0.0008	0.0008	0.0007	0.0007
-3.0	0.0013	0.0013	0.0013	0.0012	0.0012	0.0011	0.0011	0.0011	0.0010	0.0010
-2.9	0.0019	0.0018	0.0018	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
-2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
-2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
-2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
-2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
-2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
-2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
-2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
-2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
-2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
-1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
-1.8	0.0359	0.0351	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294
-1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367
-1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455
-1.5	0.0668	0.0655	0.0643	0.0630	0.0618	0.0606	0.0594	0.0582	0.0571	0.0559
-1.4	0.0808	0.0793	0.0778	0.0764	0.0749	0.0735	0.0721	0.0708	0.0694	0.0681
-1.3	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885	0.0869	0.0853	0.0838	0.0823
-1.2	0.1151	0.1131	0.1112	0.1093	0.1075	0.1056	0.1038	0.1020	0.1003	0.0985
-1.1	0.1357	0.1335	0.1314	0.1292	0.1271	0.1251	0.1230	0.1210	0.1190	0.1170
-1.0	0.1587	0.1562	0.1539	0.1515	0.1492	0.1469	0.1446	0.1423	0.1401	0.1379
-0.9	0.1841	0.1814	0.1788	0.1762	0.1736	0.1711	0.1685	0.1660	0.1635	0.1611
-0.8	0.2119	0.2090	0.2061	0.2033	0.2005	0.1977	0.1949	0.1922	0.1894	0.1867
-0.7	0.2420	0.2389	0.2358	0.2327	0.2296	0.2266	0.2236	0.2206	0.2177	0.2148
-0.6	0.2743	0.2709	0.2676	0.2643	0.2611	0.2578	0.2546	0.2514	0.2483	0.2451
-0.5	0.3085	0.3050	0.3015	0.2981	0.2946	0.2912	0.2877	0.2843	0.2810	0.2776
-0.4	0.3446	0.3409	0.3372	0.3336	0.3300	0.3264	0.3228	0.3192	0.3156	0.3121
-0.3	0.3821	0.3783	0.3745	0.3707	0.3669	0.3632	0.3594	0.3557	0.3520	0.3483
-0.2	0.4207	0.4168	0.4129	0.4090	0.4052	0.4013	0.3974	0.3936	0.3897	0.3859
-0.1	0.4602	0.4562	0.4522	0.4483	0.4443	0.4404	0.4364	0.4325	0.4286	0.4247
0.0	0.5000	0.4960	0.4920	0.4880	0.4840	0.4801	0.4761	0.4721	0.4681	0.4641
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998